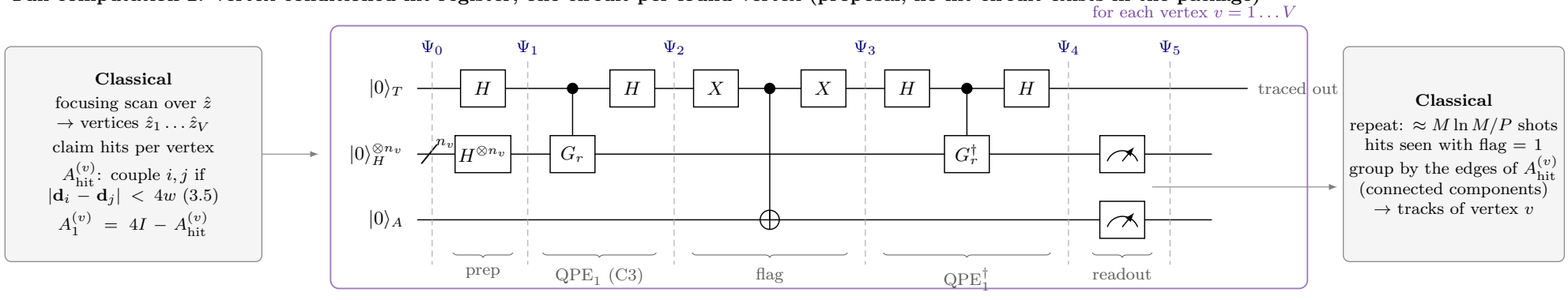
**Full computation 2: vertex-conditioned hit register, one circuit per found vertex (proposal; no hit circuit exists in the package)**




Same stages as C7 with $A \to A_1 = 4I - A_{\text{hit}}$ and $c\text{-}G \to c\text{-}G_r$ (C3). A clique $K_m$ sits on $\lambda = 5 - m$ and keeps amplitude $f_m = \cos\big(\pi(5-m)/8\big)$: $f_5 = 1$, $f_4 = 0.924$, $f_3 = 0.707$, $f_2 = 0.383$, $f_1 = 0$ (7.10).

Worked event (one vertex): $n = 4$, 6 qubits, 20 pairs. $\Psi_1$ has weight 10/16 on $\lambda = 0$ (the two $K_5$) and 6/16 on $\lambda = 4$ (noise hit and padding); $\Psi_2 = |0\rangle_A\big(\sqrt{10/16}\,|0\rangle_T|\bar{K}\rangle + \sqrt{6/16}\,|1\rangle_T|\bar{R}\rangle\big)$; ideal $P = 0.625$, 0.1 on each true hit, 0 on the noise hit.

Gates: 1033 CX at $r = 1$ ($P = 0.394$, 7.13); $r = 4$ is needed for $P$ within 3 % (4.7). Alternative filter: replace QPE$_1$, flag and QPE$_1^\dagger$ by the QSVT block of C9 on $X = (A_{\text{hit}} - 1.5)/3$ (degree-5 proposal, 8 qubits, $P = 0.441$; 7.11). Both the choice $A_1 = 4I - A_{\text{hit}}$ and the degree-5 polynomial are proposals awaiting George's agreement.